

5G RAN

Indoor Coverage Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 03 (2025-08-30).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for CSI measurement optimization by VMIMO in DAS-based indoor coverage. For details, see 4.2.1 Principles .	Modified parameter: Added the CSI_MEAS_OPT_SW option to the NRDUCellCsirs.CsIrExtSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
DAS-based indoor coverage networking	None	4.1 DAS-based Indoor Coverage Networking
VMIMO in DAS-based indoor coverage	None	4.2 VMIMO in DAS-based Indoor Coverage
Tunnel-based channel imbalance optimization in a 4T4R cell	None	5.1 Tunnel-based Channel Imbalance Optimization in a 4T4R Cell

2.3 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

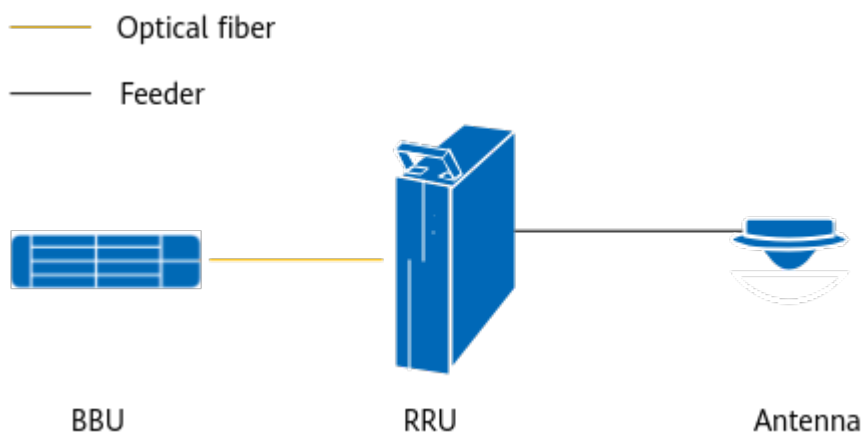
Function Name	Difference	Chapter/Section
DAS-based indoor coverage networking	Supported only in low frequency bands	4.1 DAS-based Indoor Coverage Networking
VMIMO in DAS-based indoor coverage	Supported only in low frequency bands	4.2 VMIMO in DAS-based Indoor Coverage
Tunnel-based channel imbalance optimization in a 4T4R cell	Supported only in low frequency bands	5.1 Tunnel-based Channel Imbalance Optimization in a 4T4R Cell

3 Overview

The requirements of indoor traffic density and coverage have risen rapidly as the number of mobile users in urban areas grows. Currently, approximately 70% of the traffic is generated indoors, leading to urgent demands for satisfactory indoor coverage.

Traditional macro base stations struggle to meet indoor service requirements due to drawbacks that include signal penetration loss and building blockage during indoor coverage by outdoor signals. As a result, many indoor coverage holes exist in locations such as underground parking lots, staircases, elevators, and subways. To address these issues, Huawei provides the indoor coverage solution to deploy macro base stations indoors. Indoor coverage networking generally adopts the BBU+RRU+nTnR antenna architecture, as shown in [Figure 3-1](#).

Figure 3-1 Indoor coverage networking



Indoor coverage currently applies to indoor buildings and subway tunnels. Indoor buildings mainly use distributed indoor coverage networking, which usually includes a BBU, an RRU, and a distributed antenna system (DAS). Typical networking in subway tunnel coverage scenarios includes the combination of BBU+RRU+Leaky cable. The two coverage scenarios are described in [4 DAS-based Indoor Coverage](#) and [5 Subway Tunnel Coverage](#), respectively.

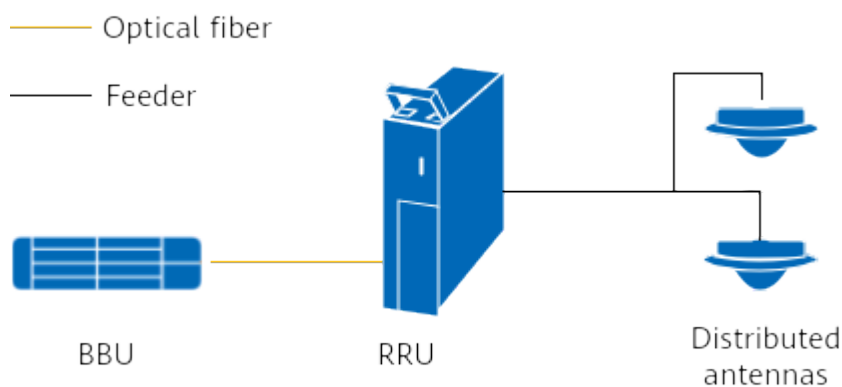
4 DAS-based Indoor Coverage

4.1 DAS-based Indoor Coverage Networking

4.1.1 Principles

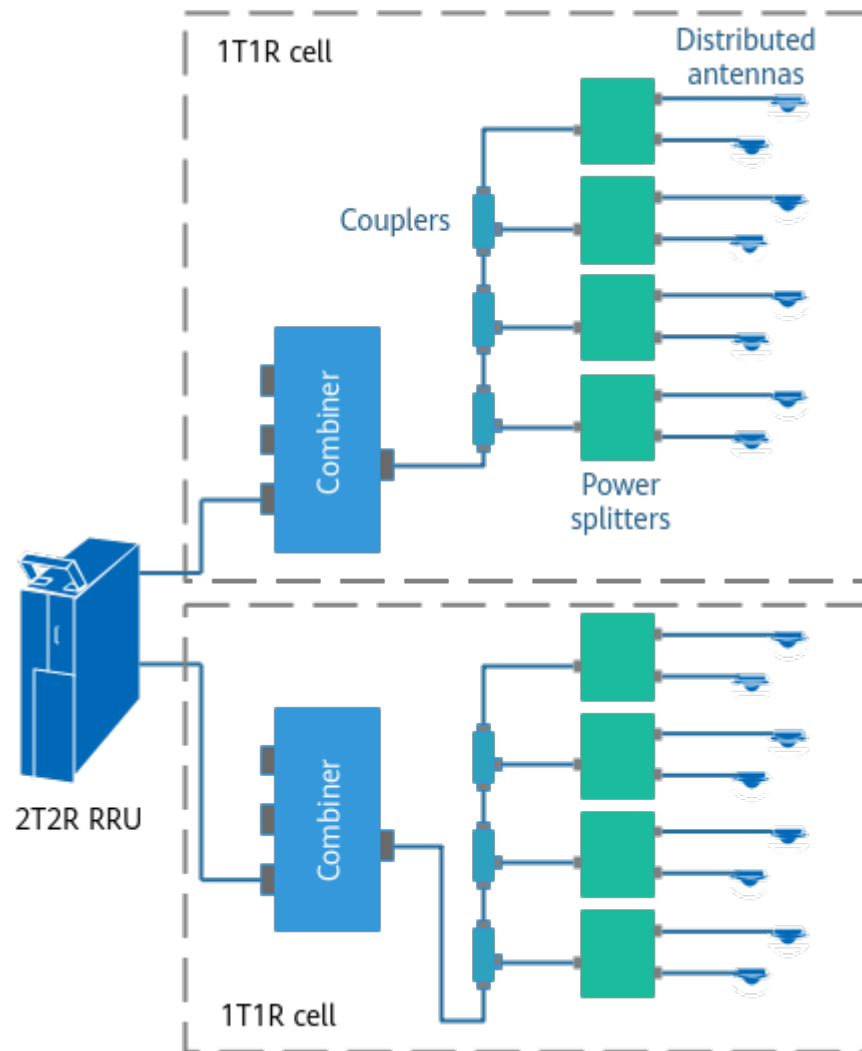
In common DAS-based indoor coverage networking, one RRU can connect to multiple distributed antennas through a split, as shown in [Figure 4-1](#).

Figure 4-1 DAS-based indoor coverage networking



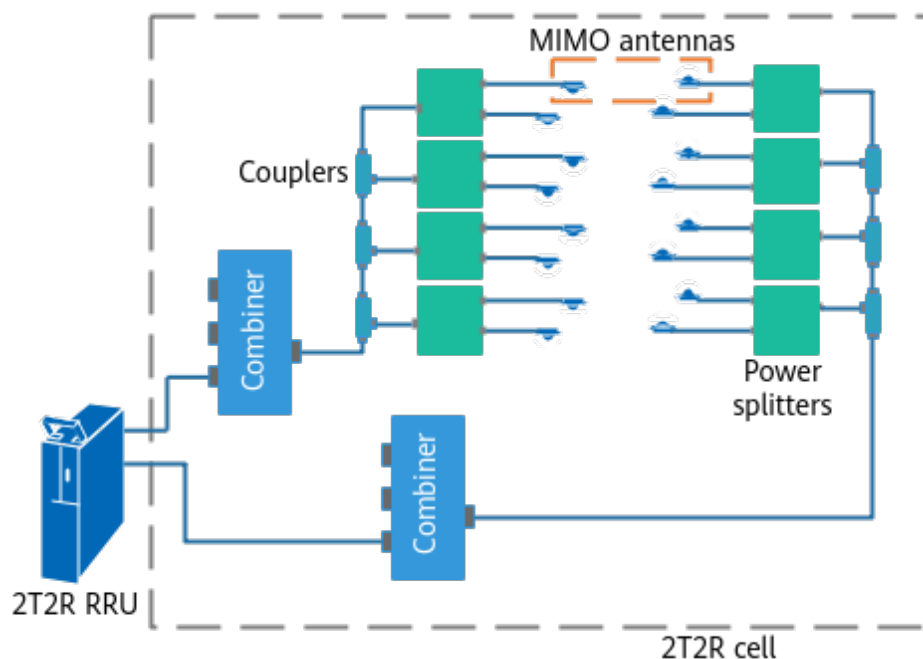
- Common DAS-based indoor coverage involves single-channel and multi-channel scenarios.
 - Single-channel scenario: For TDD, two 1T1R cells are set up using a 2T RRU through split, and the RRU connects to combiners on different floors of a building, as shown in [Figure 4-2](#).

Figure 4-2 Single-channel scenario of DAS-based indoor coverage provided by a 2T RRU



- Multi-channel scenario: For TDD, one 2T2R cell is set up using a 2T RRU that directly connects to a DAS through combiners, as shown in [Figure 4-3](#).

Figure 4-3 Multi-channel scenario of DAS-based indoor coverage provided by a 2T RRU



4.1.2 Network Analysis

4.1.2.1 Benefits

Traditional macro base stations struggle to provide excellent signal quality due to drawbacks that include signal penetration loss and building blockage during indoor coverage by outdoor signals. DAS-based indoor coverage networking expands indoor coverage and reduces coverage holes, effectively resolving issues such as poor indoor signal quality.

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

None

4.1.3 Requirements

4.1.3.1 Licenses

There are no license requirements for basic functions.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

None

4.1.3.2.2 Mutually Exclusive Functions

None

4.1.3.2.3 Function Impacts

None

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

In common DAS-based indoor coverage scenarios, all NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

In common DAS-based indoor coverage scenarios, all NR TDD-capable 2T2R RF modules except for Book RRUs support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

For details about how to set up a 2T2R cell using a 2T2R RRU, see section "Data Configuration" in *Cell Management*.

In common DAS-based indoor coverage scenarios, two 1T1R cells can be set up using a 2T RRU through split.

4.1.4.1.1 Data Preparation

Step 1 Run the **ADD RRUCHAIN** command to add an RRU chain or ring. [Table 4-1](#) describes the key parameters.

Table 4-1 Parameters in an **RRUCHAIN** MO

Parameter Name	Parameter ID	Setting Notes
Chain No.	RRUCHAIN.RCN	Set this parameter to a unique value across a base station.
Topo Type	RRUCHAIN.TT	Set this parameter as planned.
Backup Mode	RRUCHAIN.BM	Set this parameter as planned.
Access Type	RRUCHAIN.AT	Set this parameter as planned.
Protocol Type	RRUCHAIN.PROTOCOL	Set this parameter as planned.
Head Cabinet No.	RRUCHAIN.HCN	Set these parameters based on hardware installation information.
Head Subrack No.	RRUCHAIN.HSRN	
Head Slot No.	RRUCHAIN.HSN	
Head Port No.	RRUCHAIN.HPN	
Tail Cabinet No.	RRUCHAIN.TCN	
Tail Subrack No.	RRUCHAIN.TSRN	
Tail Slot No.	RRUCHAIN.TSN	
Tail Port No.	RRUCHAIN.TPN	
CPRI Line Rate	RRUCHAIN.CR	Set this parameter to a user-defined CPRI line rate or the value AUTO .

Step 2 Run the **ADD RRU** command to add an RRU. [Table 4-2](#) describes the key parameters.

Table 4-2 Parameters in an **RRU** MO

Parameter Name	Parameter ID	Setting Notes
Cabinet No.	RRU.CN	Set these parameters as planned.
Subrack No.	RRU.SRN	

Parameter Name	Parameter ID	Setting Notes
Slot No.	RRU.SV	
Topo Position	RRU.TP	
RRU Chain No.	RRU.RCN	
RRU Position	RRU.PS	
RRU type	RRU.RT	
RF Unit Working Mode	RRU.RS	
RRU Name	RRU.RN	
Number of RX channels	RRU.RXNUM	
Number of TX channels	RRU.TXNUM	
User Label	RRU.USERLABEL	
RF DC Power Off Alarm Detect Switch	RRU.RFDCPWROFFALMDETECTSW	
Maintenance Mode	RRU.MNTMODE	
DL Signal Quality Detection Switch	RRU.RFTXSIGNDETECTSW	

Step 3 Run the **ADD SECTOR** command to add a sector and a piece of sector equipment. If a 2T2R cell is set up using a 2T RRU, the procedure is the same as that for setting up a common cell. If two 1T1R cells are set up using a 2T RRU after split, the procedure is different from that for setting up common cells. For details, see the MML command example in [4.1.4.1.2 Using MML Commands](#).

Step 4 Set parameters for the cells the same way as those for common cells.

1. Run the **ADD NRDUCELL** command to add an NR DU cell.
2. Run the **ADD NRDUCELLTRP** command to add a transmission reception point (TRP) for the NR DU cell.
3. Run the **ADD NRDUCELLCOVERAGE** command to add an NR DU cell coverage item.
4. Run the **ADD NRCELL** command to add a cell.

Step 5 Run the **ACT NRCELL** command to activate the cell.

----End

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

- For details about how to set up a 2T2R cell on a 2T2R RRU, see section "Data Configuration" in *Cell Management*.
- In common DAS-based indoor coverage scenarios, two 1T1R cells can be set up on a 2T2R RRU through split. The procedure for configuring sector equipment is as follows:

```
//Adding an RRU chain
ADD RRUCHAIN: RCN=1, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=0, HSN=3, HPN=1;
//Adding an RRU
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=1, PS=0, RT=MRRU, RS=NO, RXNUM=2, TXNUM=2,
MNTMODE=NORMAL, RFDCPWROFFALMDetectSW=OFF, RFTXSIGNDetectSW=OFF;
//Adding sector 1 and sector equipment 1
ADD SECTOR: SECTORID=0, ANTNUM=1, ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A,
CREATESECTOREQM=TRUE, SECTOREQMID=0;
//Adding sector 2 and sector equipment 2
ADD SECTOR: SECTORID=1, ANTNUM=1, ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0B,
CREATESECTOREQM=TRUE, SECTOREQMID=1;
```

After sector equipment is configured, the subsequent procedure is the same as that for setting up a common cell. For operations required to set up two 1T1R cells using two pieces of sector equipment, see *Cell Management*.

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

Step 1 Run the **DSP BRD** command to check the RRU status. If the value of **Availability Status** is **Normal**, the RRU is working properly.

Step 2 Run the **DSP NRCELL** command to check the cell status. If the value of **Cell Available State** is **Available**, the cells have been activated.

----End

4.1.4.3 Network Monitoring

N/A

4.2 VMIMO in DAS-based Indoor Coverage

4.2.1 Principles

With DAS-based indoor coverage provided by a macro base station, if adjacent floors of a building are covered by different cells, interference is generated between them, affecting user experience. To mitigate interference and improve user experience, virtual multiple-input multiple-output (VMIMO) in DAS-based indoor coverage is now supported. Currently, this function applies to the following scenarios:

- Dual-channel scenarios of DAS-based indoor coverage provided by a macro base station: VMIMO can be implemented in a 4T4R cell served by combined 2T2R RRUs. **Figure 4-4** shows the networking of the two 2T2R cells serving adjacent floors, where a UE is in the overlapping area of the cells.
- Single-channel scenarios of DAS-based indoor coverage provided by a macro base station: VMIMO in a 2T2R cell is supported. **Figure 4-5** shows the networking of the two 1T1R cells serving adjacent floors, where a UE is in the overlapping area of the cells.

Figure 4-4 VMIMO in a 4T4R cell served by combined 2T2R RRUs

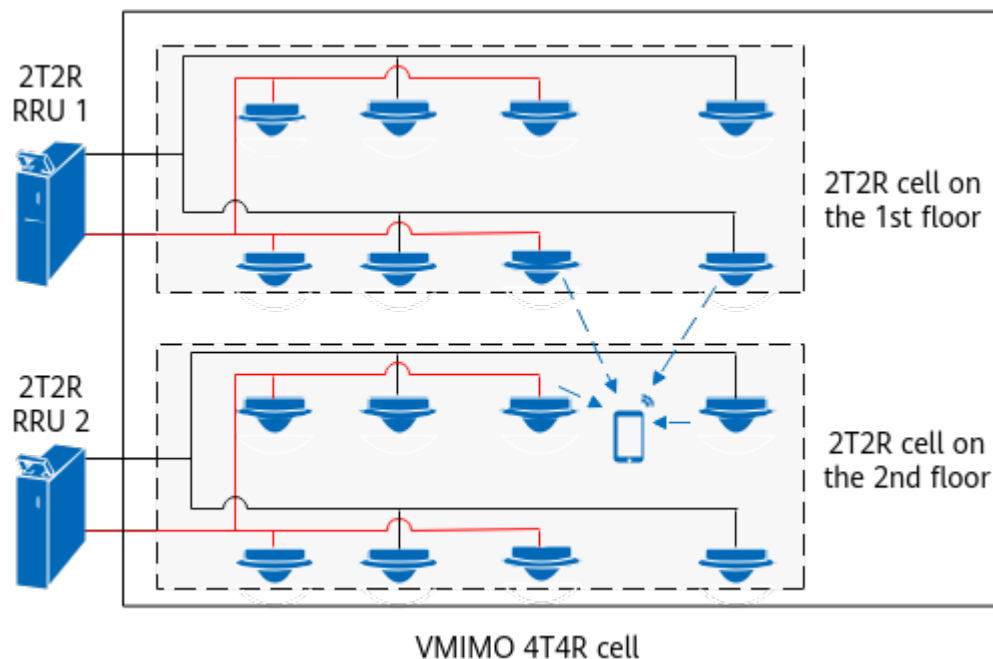
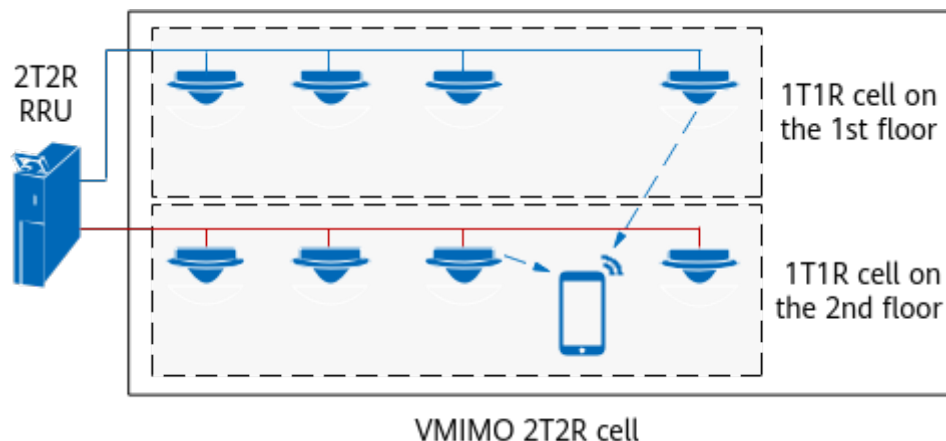


Figure 4-5 VMIMO in a 2T2R cell



The **NRDUCellAlgoSwitch.VmimoSwitch** parameter specifies whether to enable VMIMO in the preceding two scenarios. After VMIMO is enabled, the following methods are used to determine whether a UE is in the overlapping area:

- For VMIMO in a combined-RRU 4T4R cell, the gNodeB checks the difference in the SRS RSRP of a UE between RRU 1 and RRU 2. (SRS is short for sounding reference signal and RSRP is short for reference signal received power.) If the difference is small, the UE is in the overlapping area; otherwise, the UE is in the non-overlapping area.
- For VMIMO in a 2T2R cell, the gNodeB checks the difference in the SRS RSRP of a UE between the two channels of the RRU. If the difference is small, the UE is in the overlapping area; otherwise, the UE is in the non-overlapping area.

Figure 4-6 shows that the gNodeB triggers two working modes according to whether a UE is in the overlapping area.

Figure 4-6 VMIMO working modes

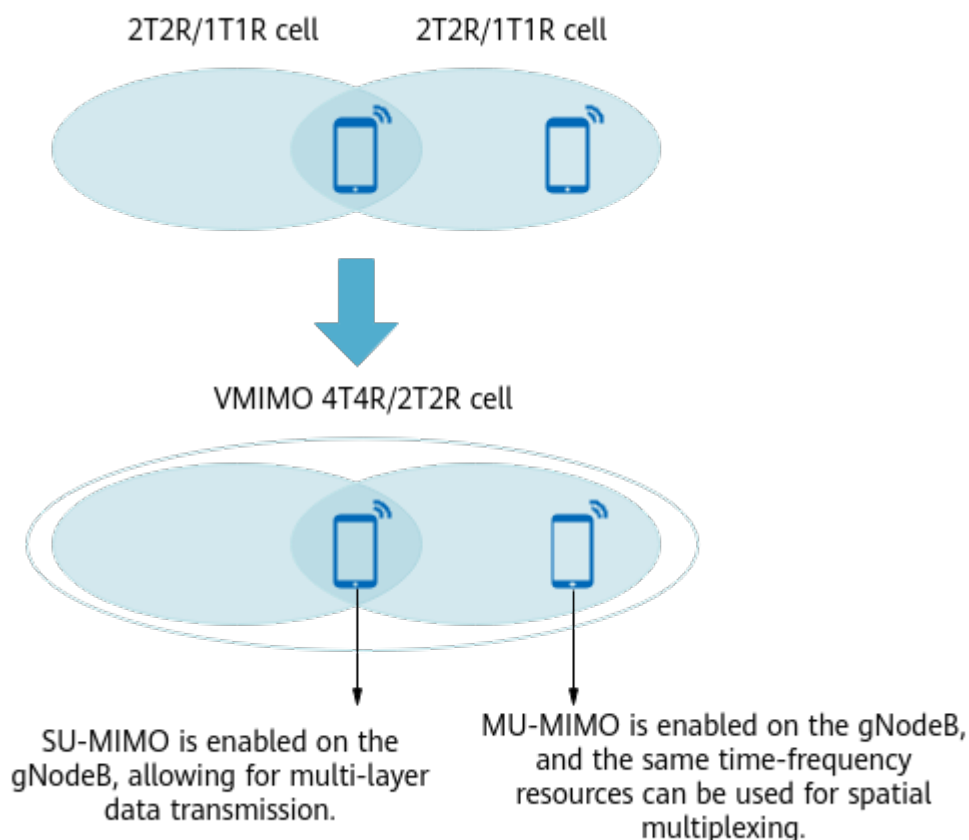


Figure 4-7 shows the working process of VMIMO. After VMIMO is enabled, the following methods are used to determine the scheduling mode:

- For VMIMO in a combined-RRU 4T4R cell, if the difference in the SRS RSRP of a UE between RRU 1 and RRU 2 is less than or equal to the value of the **NRDUCellCsirs.JtUeCsirsMeasThld** parameter, joint measurement of all channels is used for CSI-RS measurement. Otherwise, independent measurement of the RRU channel with higher SRS RSRP is used. When the SRS RSRP difference is less than or equal to the value of the **NRDUCellPdsch.VmimoDlUtSchThld** parameter, joint scheduling is used in the downlink. Otherwise, independent scheduling is used. If the SRS RSRP difference is less than or equal to the value of the **NRDUCellPusch.VmimoUlUtSchThld** parameter, joint scheduling is used in the uplink. Otherwise, independent scheduling is used.
- For VMIMO in a 2T2R cell, if a UE's SRS RSRP difference between the two channels is less than or equal to the value of the **NRDUCellCsirs.JtUeCsirsMeasThld** parameter, joint measurement of both channels is used for CSI-RS measurement. Otherwise, independent measurement of the channel with higher SRS RSRP is used. When the SRS RSRP difference is less than or equal to the value of the **NRDUCellPdsch.VmimoDlUtSchThld** parameter, joint scheduling is used in the downlink. Otherwise, independent scheduling is used. If the SRS RSRP difference is less than or equal to the value of the

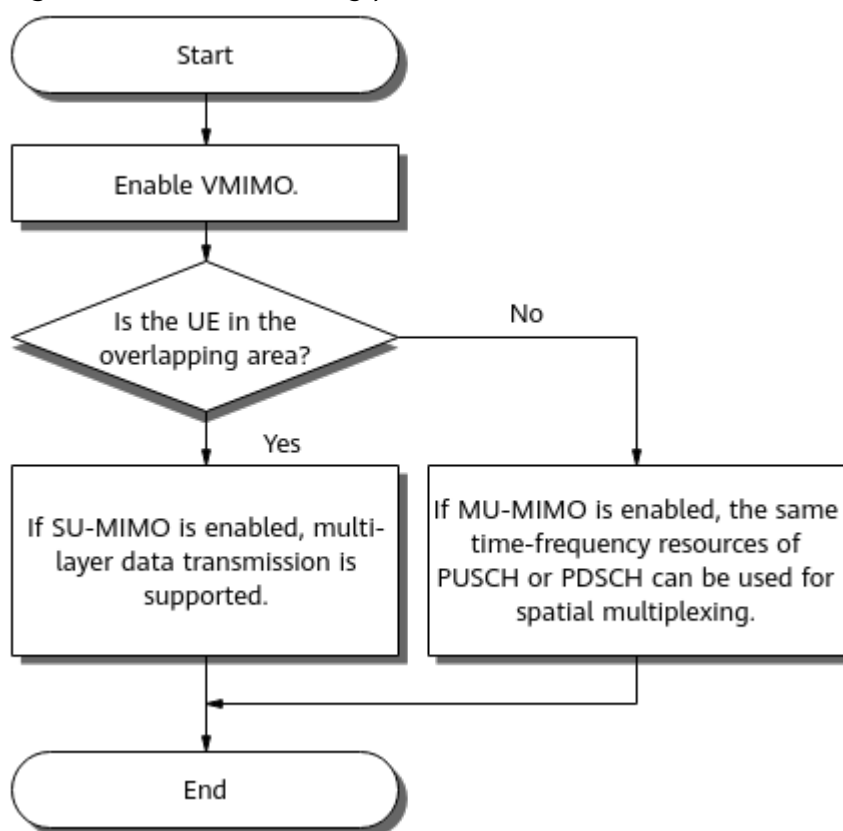
NRDUCellPusch.VmimoUUtSchThld parameter, joint scheduling is used in the uplink. Otherwise, independent scheduling is used.

After VMIMO is enabled, its performance can be improved through different functions based on UE types:

- For a UE in the overlapping area: SU-MIMO can be used, which is enabled by selecting the **UL_SU_MULTI_LAYER_SW** or **DL_SU_MULTI_LAYER_SW** option of the **NRDUCellAlgoSwitch.SuMimoMultipleLayerSw** parameter. As SU-MIMO allows for multi-layer data transmission and improves uplink or downlink capacity, enabling this function improves VMIMO performance.
- For a UE in the non-overlapping area: MU-MIMO can be used, which is enabled by selecting the **UL_MU_MIMO_SW** or **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter. As the beams can use the same time-frequency resources of PUSCH or PDSCH for spatial multiplexing, enabling MU-MIMO improves VMIMO performance.

If VMIMO in DAS-based indoor coverage is enabled and aperiodic CSI-RS resources are insufficient for UEs in non-overlapping areas, it is recommended that CSI measurement optimization be enabled to adaptively adjust the use of periodic CSI-RS resources and prevent CSI-RS measurement failures for these UEs. This optimization is controlled by the **CSI_MEAS_OPT_SW** option of the **NRDUCellCsirs.CsirsExtSwitch** parameter. For details about CSI measurement optimization, see *Channel Management*.

Figure 4-7 VMIMO working process



4.2.2 Network Analysis

4.2.2.1 Benefits

VMIMO in a 4T4R Cell Served by Combined 2T2R RRUs

- It is recommended that this function be enabled when the total downlink physical resource block (PRB) usage of common cells is less than or equal to 90%. After this function is enabled, the average throughput of UEs in the overlapping area increases by 0%–10% in the uplink and by 0%–20% in the downlink. The specific gains depend on factors such as the number of UEs in the overlapping area, the SRS RSRP difference between the two RRUs, and interference.
- PRB resources of two common 2T2R cells are halved after they are combined into a VMIMO 4T4R cell served by combined 2T2R RRUs. Therefore, it is not recommended that this function be enabled when the total downlink PRB usage of common cells is greater than 90%. If this function is enabled, PRB resources are insufficient after the combination, which may cause negative impacts. As a result, the average UE throughput and average cell throughput decrease. The specific impacts depend on factors such as the UE distribution and traffic model before the combination.

VMIMO in a 2T2R Cell

- It is recommended that this function be enabled when the total PRB usage of common cells is less than or equal to 90%. After this function is enabled, the average throughput of UEs in the overlapping area increases by 0%–20% in the uplink and by 0%–30% in the downlink. The specific gains depend on factors such as the number of UEs in the overlapping area, the SRS RSRP difference between the two channels, and interference. In single-channel scenarios of DAS-based indoor coverage provided by macro base stations, VMIMO is not recommended for a cell if the cell already operates in 2T2R mode.
- PRB resources of two common 1T1R cells are halved after the cells are combined into a VMIMO 2T2R cell. Therefore, it is not recommended that this function be enabled when the total downlink PRB usage of common cells is greater than 90%. If this function is enabled, PRB resources are insufficient after the combination, which may cause negative impacts. As a result, the average UE throughput and average cell throughput decrease. The specific impacts depend on factors such as the UE distribution and traffic model before the combination.

NOTE

The average uplink UE throughput is indicated by **User Uplink Average Throughput (DU)**.
The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**.
The downlink PRB usage is indicated by **Downlink Resource Block Utilizing Rate (DU)**.

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

- VMIMO in a 4T4R cell served by combined 2T2R RRUs
 - In NR TDD, with this function enabled, if one of the combined 2T2R RRUs serving a VMIMO 4T4R cell becomes faulty, joint data transmission cannot be implemented in the overlapping area. In this case, the cell is considered faulty, that is, the cell is unavailable.
 - In NR TDD, the synchronization signal and PBCH block (SSB) beams consume more resources (the resource loss is about 2%) because the cell configuration changes from 2T2R to 4T4R. In addition, the PDCCH CCE aggregation level and CCE usage may increase.
 - After VMIMO is enabled, a set of resources for aperiodic CSI-RS measurements is added, which may slightly decrease the average downlink MCS index.
- VMIMO in a 2T2R cell
 - In NR TDD, with this function enabled, if one channel of a VMIMO 2T2R cell becomes faulty, joint data transmission cannot be implemented in the overlapping area. In this case, the cell is considered faulty, that is, the cell is unavailable.
 - In NR TDD, the SSB beams consume more resources (the resource loss is about 1%) because the cell configuration changes from 1T1R to 2T2R. In addition, the PDCCH CCE aggregation level and CCE usage may increase.
 - After VMIMO is enabled, a set of resources for aperiodic CSI-RS measurements is added, which may slightly decrease the average downlink MCS index.

4.2.3 Requirements

4.2.3.1 Licenses

There are no license requirements.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

None

4.2.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	None
TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	None
TDD	LTE FDD and NR Uplink Spectrum Sharing	Options of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter: • LTE_NR_UL_SPECTRUM_SHARING_SW • LTE_NR_FDD_SPCT_SHR_SW	<i>LTE FDD and NR SUL Dynamic Spectrum Sharing</i>	None
TDD	Remote interference adaptive avoidance	RIM_ADAPT_AVOID_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	None
TDD	Enhanced remote interference adaptive avoidance	RIM_ADAPT_AVOID_ENH_SW option of the NRDUCellAlgoSwitch.RimAlgoSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	None
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	None
Low-frequency TDD	3GPP R15-compliant uplink slot aggregation	VONR_UL_SLOT_AGGREGATION_SW option of the NRDUCellUISch.VoiceULSchSwitch parameter	<i>VoNR</i>	None

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	None
Low-frequency TDD	D-MIMO Service Switch	D_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None	None
Low-frequency TDD	DL 1T4R UE SRS Period Shorten Switch	DL_1T4R_SRS_PERIOD_SHORTEN_SW option of the NRDUCellPdschPre-code.SuMimoSrsPeriodShortSw parameter	None	None
Low-frequency TDD	TRP Antenna Type	NRDUCellTrpBeam.TrpAntType set to 8T_2222 , 8T_44 , or 8T_422 .	None	None
Low-frequency TDD	Common-channel dynamic shutdown in TTI-level channel shutdown	TTI_RF_SHUTDOWN_SWITCH option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter selected and NRDUCellPowerSaving.ComChnDynShutdownSwitch set to ON	<i>Energy Conservation and Emission Reduction</i>	None
Low-frequency TDD	pRRU-level precise symbol power saving	DL_PRECISE_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	None

RAT	Function Name	Function Switch	Reference	Description
TDD	Common-channel dynamic shutdown	NRDUCellPowerSaving.ComChnDynShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	When VMIMO in DAS-based indoor coverage is enabled in a cell and an NRDUCellPowerSavingMO with NRDUCellPowerSaving.PowerSavingType set to TTI_RF_SHUTDOWN or RF_CHN_DYNAMIC_SHUTDOWN is configured for the cell, the switch for common-channel dynamic shutdown must be set to OFF .
Low-frequency TDD	3GPP R16-compliant uplink slot aggregation	VOICE_R16_UL_SLOT_AGG_SW option of the NRDUCellUISch.VoiceULSchSwitch parameter	<i>VoNR</i>	None

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell-level PUSCH interference avoidance	- Both the UL_CELL_LEVEL_INTRF_AVOID and DL_UE_LEVEL_INTRF_AVOID options of the NRDUCellIntrfIdent./ntrfOptimizationMethod parameter selected NRDUCellIntrfIdent./ntrfIdentificationMethod set to UL_DL_MANUAL - At least one uplink interference range configured (using the NRDUCellIntrfBand MO with NRDUCellIntrfBand./nterferenceDirection set to UPLINK)	<i>Interference Avoidance</i>	None
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTRF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	VMIMO in DAS-based indoor coverage cannot be enabled together with inter-cell interference avoidance.

4.2.3.2.3 Function Impacts

RA T	Function Name	Function Switch	Reference	Description
Lo w- freq uen cy TDD	MU-MIMO (low- frequency TDD)	NRDUCellAlgoSwitch.MuMimoSwitch	<i>MIMO (TDD)</i>	For UEs in the non-overlapping areas, if the UL_MU_MIMO_SW or DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter is selected, the beams can use the same time-frequency resources of PUSCH or PDSCH for spatial multiplexing.
Lo w- freq uen cy TDD	Cell load simulation	STRNRDUCELLSIMULATELOAD	None	Cell load simulation cannot be enabled when VMIMO is enabled. If cell load simulation has been enabled, it will be disabled as soon as VMIMO is enabled.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards as well as NR TDD-capable UBBPg and later boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR 2T2R RF modules except for Book RRUs support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Others

VMIMO in a 4T4R Cell Served by Combined 2T2R RRUs

- The two 2T2R RRUs serving a 4T4R cell must be of the same model and operate in the same frequency band.
- The two RRUs can connect to the same baseband processing unit or different baseband processing units in a BBU, but cannot connect to different BBUs.
- The difference between BBU-RRU optical fibers cannot exceed 2 km.

VMIMO in a 2T2R Cell

None

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

[Table 4-3](#) and [Table 4-4](#) describe the parameters used for function activation and optimization, respectively. This section does not describe parameters related to cell establishment.

Table 4-3 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
VMIMO Switch	NRDUCellAlgoSwitch.VmimoSwitch	Set this parameter to ON if VMIMO is required.
Joint UE CSI-RS Measurement Threshold	NRDUCellCsirs.JtUeCsirsMeasThld	For VMIMO in a 4T4R cell served by combined 2T2R RRUs, the value 15 is recommended. For VMIMO in a 2T2R cell, the value 20 is recommended.
VMIMO DL Joint Scheduling Threshold	NRDUCellPdsch.VmimoDLJtSchThld	For VMIMO in a 4T4R cell served by combined 2T2R RRUs, the value 20 is recommended. For VMIMO in a 2T2R cell, the value 25 is recommended.

Parameter Name	Parameter ID	Setting Notes
VMIMO UL Joint Scheduling Threshold	NRDUCellPusch.VmimoULtSchThld	For VMIMO in a 4T4R cell served by combined 2T2R RRUs, the value 20 is recommended. For VMIMO in a 2T2R cell, the value 25 is recommended.

 NOTE

The recommended values for the thresholds listed in [Table 4-3](#) are based on the SSB RSRP values ranging from -80 dBm to -90 dBm. If the actual SSB RSRP is higher, the thresholds can be raised for optimization. If the actual SSB RSRP is lower, the thresholds can be lowered.

Table 4-4 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
SU-MIMO Multiple Layers Switch	NRDUCellAlgoSwitch.SuMimoMultipleLayerSw	Select the UL_SU_MULTI_LAYER_SW or DL_SU_MULTI_LAYER_SW option when VMIMO is enabled.
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	Select the UL_MU_MIMO_SW or DL_MU_MIMO_SW option when VMIMO is enabled.
Uplink PUSCH Algorithm Switch	NRDUCellPusch.UlPuschAlgoSwitch	Select the UL_SCHEDULE_ENH_SW option when both VMIMO and uplink MU-MIMO are enabled.
UL MU-MIMO Algorithm Switch	NRDUCellUlMimo.UlMuMimoAlgoSwitch	Select the MU_MULTI_SLOT_PAIR_SW option when both VMIMO and uplink MU-MIMO are enabled.
Uplink Inconsecutive Scheduling Switch	NRDUCellAlgoSwitch.UlInconsecutiveSchSwitch	Deselect the UL_NON_CON_SCH_SW option when both VMIMO and uplink MU-MIMO are enabled.

Parameter Name	Parameter ID	Setting Notes
UL MU-MIMO Algorithm Switch	NRDUCellUIMimo.UIMuMimoAlgoSwitch	Select the DIFF_FLDMRS_MAX_LENGTH_PAIR_SW option when both VMIMO and uplink MU-MIMO are enabled.

 NOTE

For details about how to enable SU-MIMO and MU-MIMO, see *MIMO*.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

The **NRDUCellAlgoSwitch.VmimoSwitch** parameter is a high-risk parameter. Modifying this parameter will cause the cell to automatically restart.

Activation Command Examples

For the command examples for a combined-RRU cell, see the descriptions of combined-RRU cell establishment in the "Using MML Commands" section of *Cell Management*. The following configurations only describe the difference for activating VMIMO.

Configuring VMIMO in a 4T4R cell served by combined 2T2R RRUs:

```
//Adding a sector
ADD SECTOR: SECTORID=0, ANTNUM=4, ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANT2CN=0,
ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=61, ANT3SN=0, ANT3N=R0A, ANT4CN=0,
ANT4SRN=61, ANT4SN=0, ANT4N=R0B, CREATESECTOREQM=FALSE;
//Setting the sector equipment
ADD SECTOREQM: SECTOREQMID=0, SECTORID=0, ANTFCGMODE=ANTENNAPORT, ANTNUM=4,
ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANTTYPE1=RXTX_MODE, ANT2CN=0, ANT2SRN=60,
ANT2SN=0, ANT2N=R0B, ANTTYPE2=RXTX_MODE, ANT3CN=0, ANT3SRN=61, ANT3SN=0, ANT3N=R0A,
ANTTYPE3=RXTX_MODE, ANT4CN=0, ANT4SRN=61, ANT4SN=0, ANT4N=R0B, ANTTYPE4=RXTX_MODE;
//Enabling VMIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, VmimoSwitch=ON;
MOD NRDUCELLCSIRS: NrDuCellId=0, JtUeCsirsMeasThld=15;
MOD NRDUCELLPDSCH: NrDuCellId=0, VmimoDUtSchThld=20;
MOD NRDUCELLPUSCH: NrDuCellId=0, VmimoUUtSchThld=20;
```

Configuring VMIMO in a 2T2R cell:

```
//Adding a sector
ADD SECTOR: SECTORID=0, ANTNUM=2, ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANT2CN=0,
ANT2SRN=60, ANT2SN=0, ANT2N=R0B, CREATESECTOREQM=FALSE;
//Setting the sector equipment
ADD SECTOREQM: SECTOREQMID=0, SECTORID=0, ANTCFGMODE=ANTENNAPORT, ANTNUM=2,
ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANTTYPE1=RX_TX_MODE, ANT2CN=0, ANT2SRN=60,
ANT2SN=0, ANT2N=R0B, ANTTYPE2=RX_TX_MODE;
//Enabling VMIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, VmimoSwitch=ON;
MOD NRDUCELLCSIRS: NrDuCellId=0, JtUeCsirsMeasThld=20;
MOD NRDUCELLPDSCH: NrDuCellId=0, VmimoDUtSchThld=25;
MOD NRDUCELLPUSCH: NrDuCellId=0, VmimoUUtSchThld=25;
```

Optimization Command Examples

The optimization configurations are the same in different scenarios.

```
//Enabling uplink SU-MIMO and downlink SU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
SuMimoMultipleLayerSw=UL_SU_MULTI_LAYER_SW-1&DL_SU_MULTI_LAYER_SW-1;
//Enabling uplink MU-MIMO and downlink MU-MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=UL_MU_MIMO_SW-1&DL_MU_MIMO_SW-1;
//Enabling uplink scheduling enhancement
MOD NRDUCELLPUSCH: NrDuCellId=0, UIPuschAlgoSwitch=UL_SCHEDULE_ENH_SW-1;
//Enabling MU-MIMO cross-slot pairing
MOD NRDUCELLULMIMO: NrDuCellId=70, UIMuMimoAlgoSwitch=MU_MULTI_SLOT_PAIR_SW-1;
//Disabling uplink non-consecutive scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=70, UlinconsecutiveSchSwitch=UL_NON_CON_SCH_SW-0;
//Enabling UEs supporting different maximum numbers of front-loaded DMRS symbols to be paired
MOD NRDUCELLULMIMO: NrDuCellId=70, UIMuMimoAlgoSwitch=DIFF_FLDMRS_MAX_LENGTH_PAIR_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

The deactivation configurations are the same in different scenarios.

```
//Disabling VMIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, VmimoSwitch=OFF;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

Step 1 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR > Application Layer > Uu Interface Trace**.

Step 3 If the *RRC_RECFG* > *csi-MeasConfig* > *CSI-ReportConfig* signaling indicates aperiodic CSI reporting, VMIMO in DAS-based indoor coverage has taken effect.

----End

4.2.4.3 Network Monitoring

Checking the Average UE Throughput

Observe the following KPIs of the original two cells and the cell with VMIMO in DAS-based coverage enabled to check the increase in the average UE throughput.

- **User Uplink Average Throughput (DU)**
- **User Downlink Average Throughput (DU)**

Checking the Single-UE Throughput

Select UEs in the overlapping area to estimate an increase in the single-UE throughput.

- To compare the uplink MAC throughput before and after the function is enabled, choose **User Performance Monitoring** > **User Common Monitoring** > **Uplink MAC Throughput** on the MAE-Access.
- To compare the downlink MAC throughput before and after the function is enabled, choose **User Performance Monitoring** > **User Common Monitoring** > **Downlink MAC Throughput** on the MAE-Access.

NOTE

In the **User Common Monitoring** dialog box, you are advised to set **UEID Type** to **Random Value** or **STMSI** in NSA networking, and set **UEID Type** to **5G-Random Value** or **5G-STMSI** in SA networking.

5 Subway Tunnel Coverage

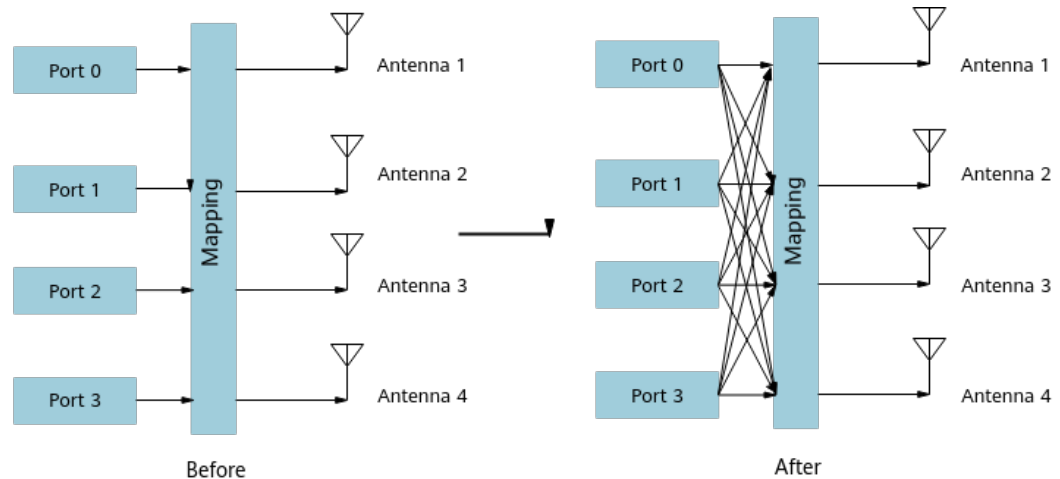
Tunnels are typically pipe-shaped, which are long and narrow. As signals are easily blocked by things like trees and mountains, coverage holes are likely to occur in tunnels. For long and curved tunnels, leaky feeder cables are mainly used for deployment, which differs from the traditional DAS-based indoor coverage solution. In cases where leaky feeder cables are used, the used cables may be of different types, resulting in different cable losses. To address this issue, tunnel-based channel imbalance optimization in a 4T4R cell is introduced. For details, see [5.1 Tunnel-based Channel Imbalance Optimization in a 4T4R Cell](#).

5.1 Tunnel-based Channel Imbalance Optimization in a 4T4R Cell

5.1.1 Principles

For a 4T4R cell along a tunnel, the loss of a 13/8" leaky feeder cable is different from that of a 5/4" leaky feeder cable. This leads to imbalanced channel signals in the cell, and therefore imbalanced power between CSI-RS ports. If there is a big difference in the CSI-RS port power, the rank indication (RI), precoding matrix indication (PMI), and channel quality indicator (CQI) may not be valid. To solve this problem, tunnel-based channel imbalance optimization in a 4T4R cell is introduced, as shown in [Figure 5-1](#). This solution changes the mapping between CSI-RS ports and TX antenna ports from one-to-one to many-to-many. When channel signals are imbalanced, many-to-many mapping balances the received CSI-RS port power between TX antennas, helping UEs report high RIs.

Figure 5-1 Port mapping before and after balancing



This function is controlled by the **CSIRS_BEAM_POWER_BALANCE_SW** option of the **NRDUCellAlgoSwitch.BeamOptSwitch** parameter.

After tunnel-based channel imbalance optimization is enabled, each physical antenna transmits data from four CSI-RS ports. Since the data from the four CSI-RS ports is orthogonal, the power between CSI-RS ports is balanced, increasing the data rates in imbalanced scenarios.

5.1.2 Network Analysis

5.1.2.1 Benefits

After tunnel-based channel imbalance optimization is enabled, the average downlink UE throughput increases by 0%–8%.

- The average downlink throughput increases by a maximum of 8% when the power difference between two channels is greater than or equal to 15 dB.
- The average downlink throughput does not increase when two channels are balanced.

NOTE

UEs using SRS-based weights cannot benefit from tunnel-based channel imbalance optimization.

The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

In NR TDD, with this function enabled, if one channel of a 4T4R cell becomes faulty, joint data transmission cannot be implemented in the overlapping area. In this case, the cell is considered faulty, that is, the cell is unavailable.

5.1.3 Requirements

5.1.3.1 Licenses

None

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

None

5.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	Intra-base-station DL CoMP does not work with tunnel-based channel imbalance optimization.
Low-frequency TDD	Inter-gNodeB DL Joint Transmission Sw	INTER_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	None	Inter-gNodeB DL Joint Transmission Sw does not work with tunnel-based channel imbalance optimization.
Low-frequency TDD	RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Tunnel-based channel imbalance optimization does not work with RF channel intelligent shutdown.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_C OMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM does not work with tunnel-based channel imbalance optimization.

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
TDD	VMIMO in a 4T4R cell served by combined RRUs	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	When VMIMO in a 4T4R cell served by combined RRUs and tunnel-based channel imbalance optimization in a 4T4R cell are both enabled, only the former function takes effect.

5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards as well as NR TDD-capable UBBPfw, UBBPg, and later boards support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR 4T RF modules except for Book RRUs support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.1.3.4 Cells

Tunnel-based channel imbalance optimization can be enabled only when the **NRDUCellTrp.TrpType** parameter is set to **DEFAULT**.

5.1.3.5 Others

None

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

5.1.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Beam Optimization Switch	NRDUCellAlgoSwitch.BeamOptSwitch	CSIRS_BEAM_POWER_BALANCE_SW	Select this option to enable tunnel-based channel imbalance optimization.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to **5.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Turning on the switch for tunnel-based channel imbalance optimization  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, BeamOptSwitch=CSIRS_BEAM_POWER_BALANCE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switch for tunnel-based channel imbalance optimization  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, BeamOptSwitch=CSIRS_BEAM_POWER_BALANCE_SW-0;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

Observe the function switch by running the **LST NRDUCELLALGOSWITCH** command. If the following information is displayed in the command output, this function has been enabled.

```
Beam Optimization Switch = CSI-RS Beam Power Balance Sw:On
```

5.1.4.3 Network Monitoring

Checking the Average UE Throughput

Check the following key performance indicator (KPI) before and after tunnel-based channel imbalance optimization is enabled to observe the increase in the average UE throughput in a cell.

The average downlink UE throughput is indicated by **User Downlink Average Throughput (DU)**.

Checking the Single-UE Throughput

Observe the increase in the throughput of a UE in the overlapping area.

To compare the downlink MAC throughput before and after the function is enabled, choose **User Performance Monitoring > User Common Monitoring > Downlink MAC Throughput** on the MAE-Access.

NOTE

In the **User Common Monitoring** dialog box, you are advised to set **UEID Type** to **Random Value** or **STMSI** in NSA networking, and set **UEID Type** to **5G-Random Value** or **5G-STMSI** in SA networking.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.
- End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Cell Management*
- *CoMP*
- *Energy Conservation and Emission Reduction*
- *LTE FDD and NR SUL Dynamic Spectrum Sharing*
- *Remote Interference Management (Low-Frequency TDD)*
- *Interference Avoidance*
- *MIMO (TDD)*
- *MIMO*
- *VoNR*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Co-MM (Low-Frequency TDD)*
- *Massive MIMO iBeam (Low-Frequency TDD)*